

QUBIT 2026 HACKATHON · CHALLENGE: ANTENNA TILT

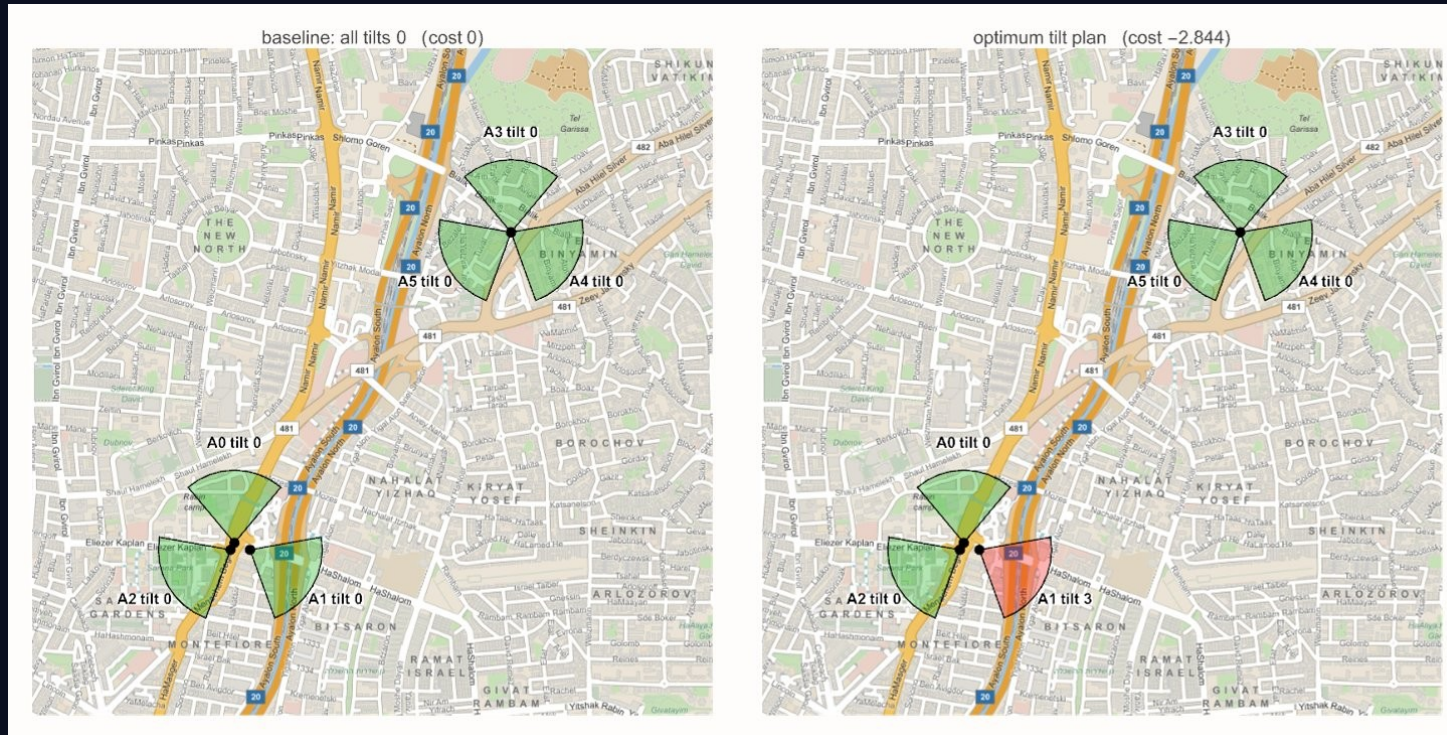
Tilting 6 Tel Aviv antennas with a 12-qubit QAOA

Real OpenCellID data → cost Hamiltonian → Classiq circuit → Wolfram exact simulation → Lean 4 proof

Team Quantum Headache

5-minute demo · September 2026

The problem: pick one down-tilt per antenna



Two real 3-sector sites from OpenCellID (Ayalon, Tel Aviv). Left: all tilts 0. Right: optimum plan — only A1 tilts to level 3.

- ▶ 6 sector antennas A0...A5, 4 tilt levels each $\rightarrow 4^6 = 4096$ plans
- ▶ Tilting down: less interference on neighbours, less coverage
- ▶ $\text{Cost}(t) = t^T I t - 1.5 \cdot c \cdot t + 0.3 \cdot \Sigma t$ (interference – coverage + penalty)
- ▶ Classical brute force: unique optimum -2.844 at $(0,3,0,0,0,0)$
- ▶ Real networks: 100s of antennas $\times$ 10 levels $\rightarrow$ intractable $\rightarrow$ quantum

Encoding: 2 qubits per antenna, 12 qubits total

$$\text{tilt}_i = q_{2i} + 2 \cdot q_{2i+1} \in \{0, 1, 2, 3\}$$

$q_i = (1 - Z_i)/2 \rightarrow$ Cost becomes a Pauli-Z polynomial

$$H_C = c_0 \cdot I + \sum h_i Z_i + \sum J_{ij} Z_i Z_j$$

73 terms: 1 identity + 12 Z + 60 ZZ — strictly 2-local, no ZZZ

- ▶ Why binary (2 qubits) and not one-hot (4 qubits) or a dense 5-qubit pack?
- ▶ Binary keeps H_C quadratic $\rightarrow$ only RZ and CX gates after ZZ decomposition
- ▶ Fits Classiq student edition comfortably; 15 parallel jobs used for sweeps
- ▶ QAOA layer: phase $\exp(+i\gamma H_C)$ then mixer $RX(2\beta)$ on every qubit
- ▶ Verified by Walsh–Hadamard: Pauli table reproduces all 4096 costs exactly

12

qubits

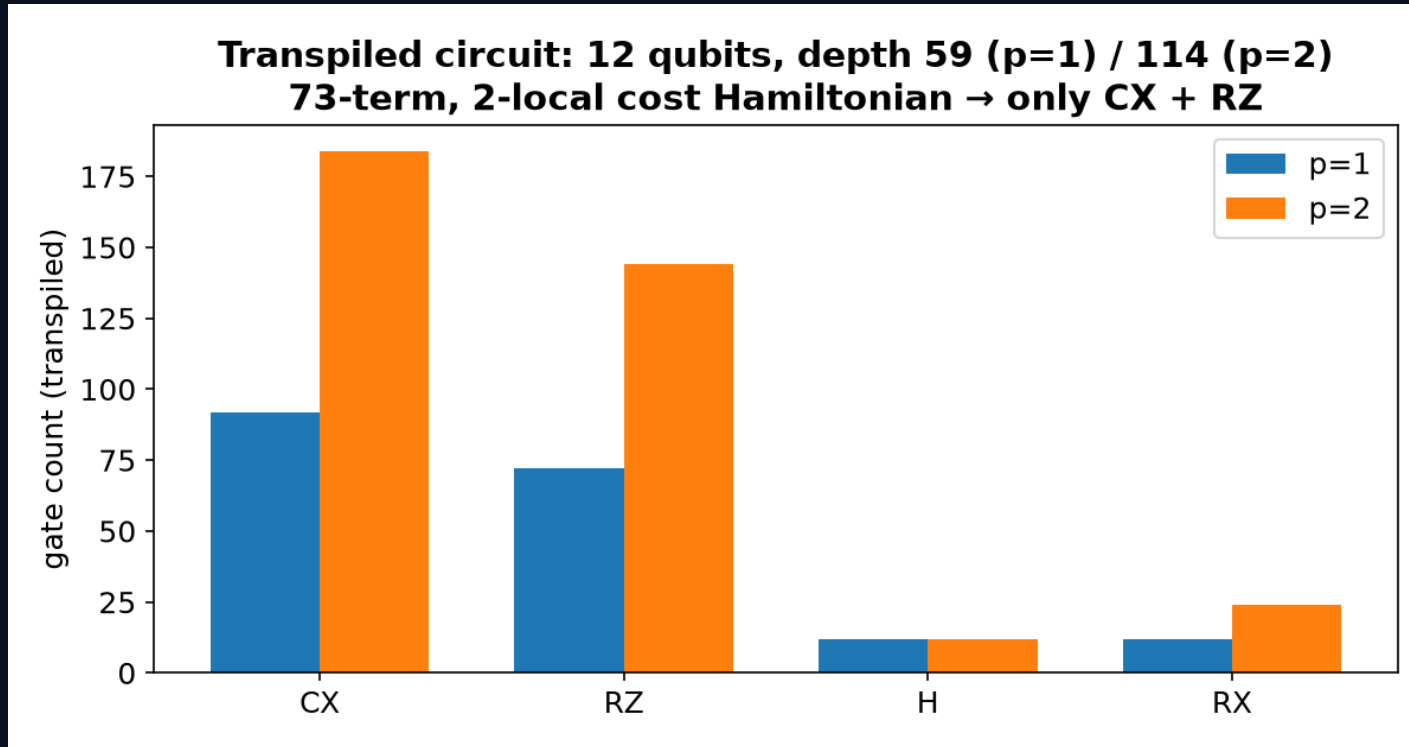
73

Pauli terms

2-local

ZZ max weight

The circuit Classiq synthesised

**59**depth, $p = 1$ **114**depth, $p = 2$ **92 CX**

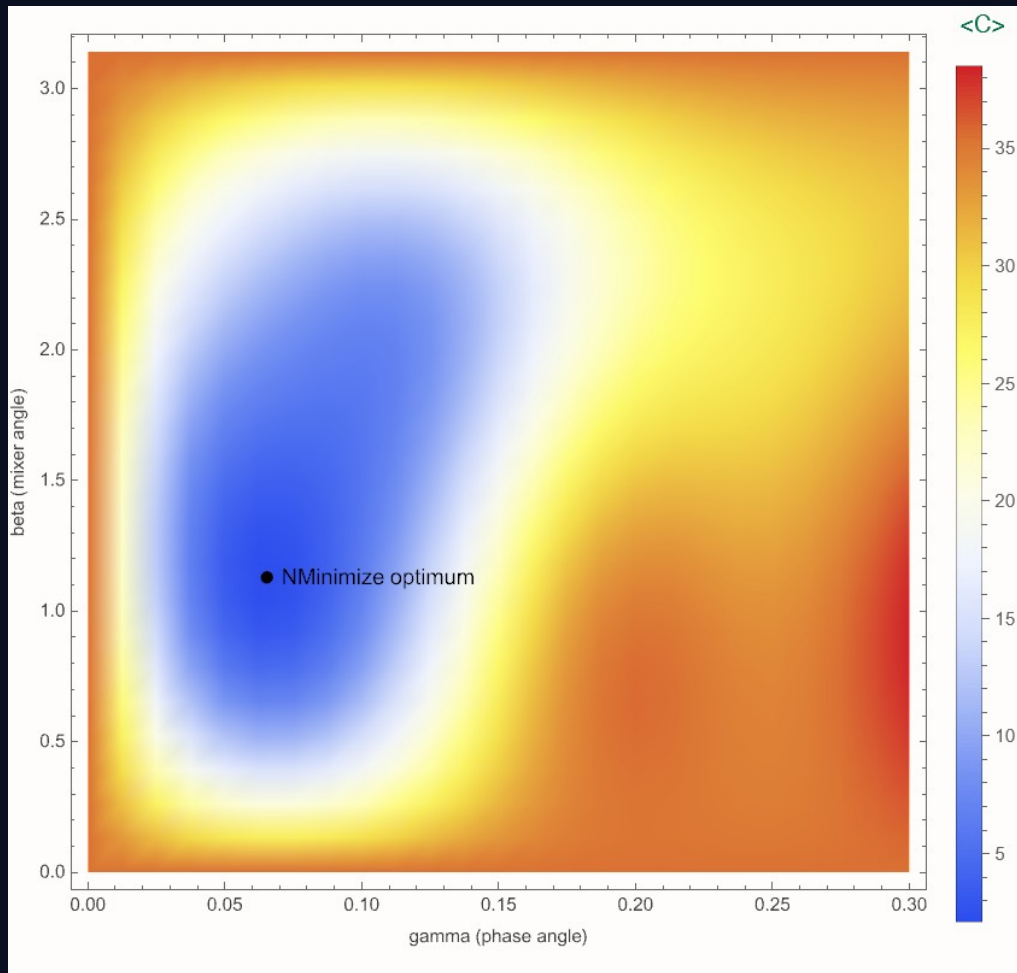
+ 72 RZ + 12 H + 12 RX

184 CX

+ 144 RZ + 12 H + 24 RX

- ▶ Written in Qmod: QStruct of QArray[QNum[2], 6], phase(cost, γ), RX mixer
- ▶ Exported as OpenQASM 3 (group/classiq/results/*.qasm) + hand-written QASM 2 layer
- ▶ Same circuit re-simulated in Qiskit statevector: $\langle C \rangle = 1.9985$ — identical to Wolfram

Wolfram exact simulation: where the good angles are



$(0.0654, 1.1293)$

optimal (γ, β) , $p = 1$

$\langle C \rangle = 1.9985$

$p = 1$ (random: 35.5)

$\langle C \rangle = -0.3825$

$p = 2, 4$ angles

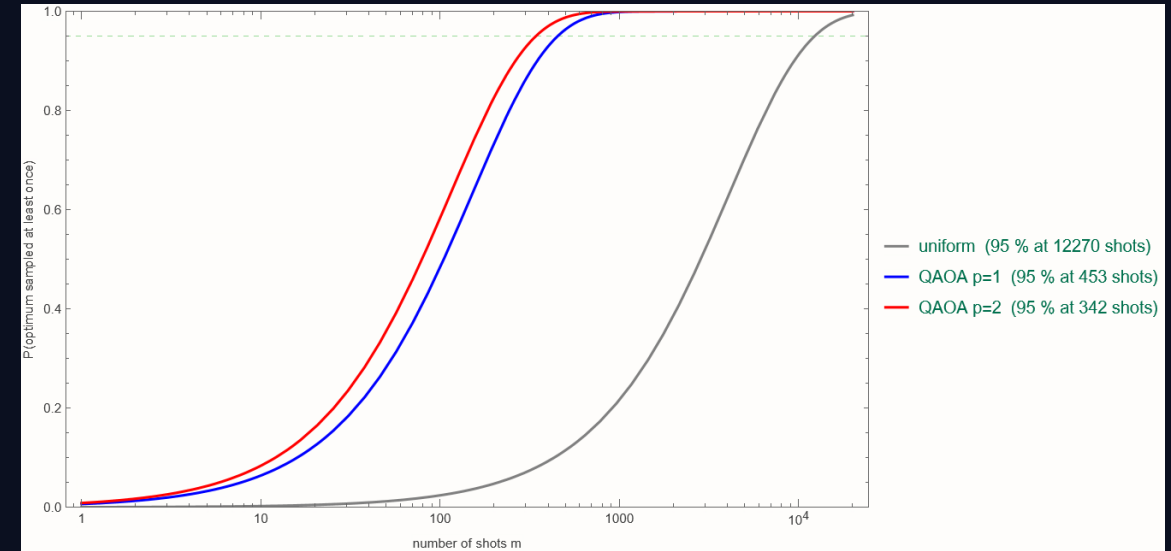
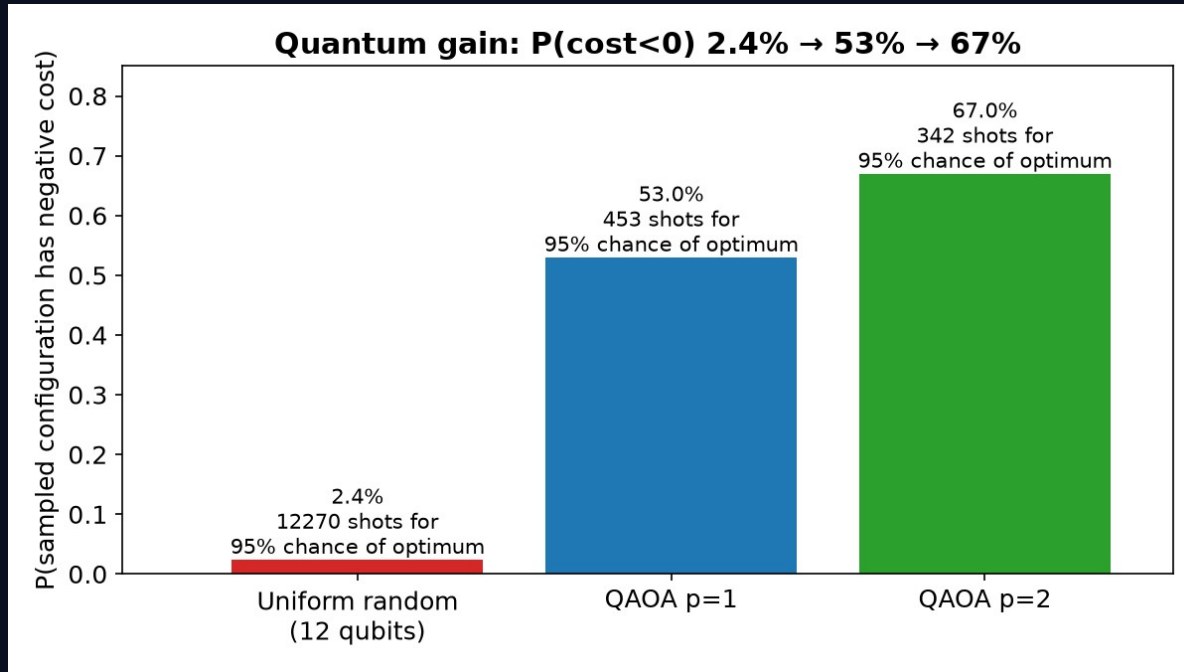
-2.844

ground state = classical optimum

- ▶ Notebook + PDF + CDF + Markdown in [group/wolfram/](#) (14 pages, 13 figures)
- ▶ Landscape is flat almost everywhere — a bad start sits on a plateau at $\langle C \rangle \approx 35$
- ▶ Angles found here are the warm start we hand to Classiq
- ▶ Also in the notebook: VQE (RY+CZ), Grover on a 2-antenna slice, 10-level scaling

$\langle C \rangle(\gamma, \beta)$ for $p = 1$ over all 4096 amplitudes (statevector, no shot noise).

What the quantum sampler buys us



Shots needed for 95% chance of sampling the exact optimum (Wolfram).

2.4% → 67%

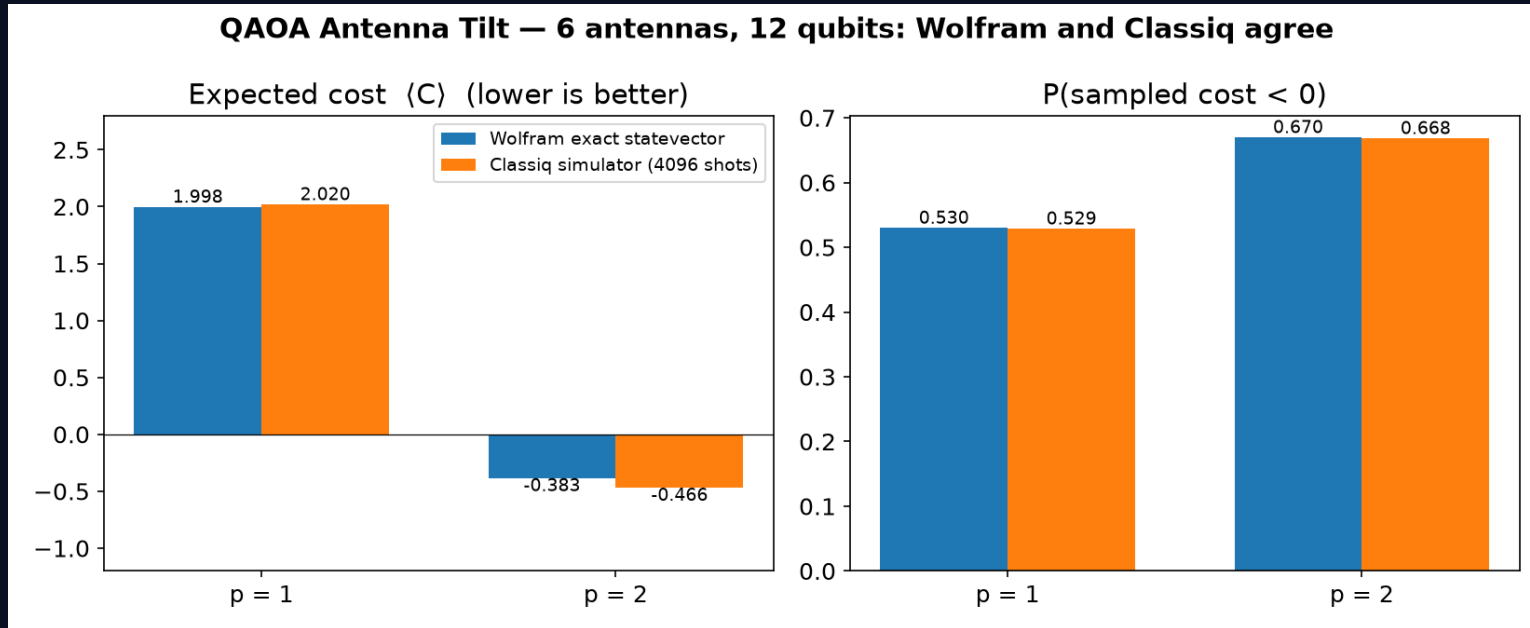
$P(\text{cost} < 0)$, random → $p = 2$

12270 → 342

shots to hit optimum (95%)

- ▶ Half of all samples at $p = 1$ and two-thirds at $p = 2$ are already better than the baseline
- ▶ Only 1–2 QAOA layers; more layers or larger p continue to sharpen the distribution

Same numbers from the Classiq simulator



2.0203 vs 1.9985

$\langle C \rangle$ p = 1, 4096 shots vs exact

0.529 vs 0.530

P(cost < 0), p = 1

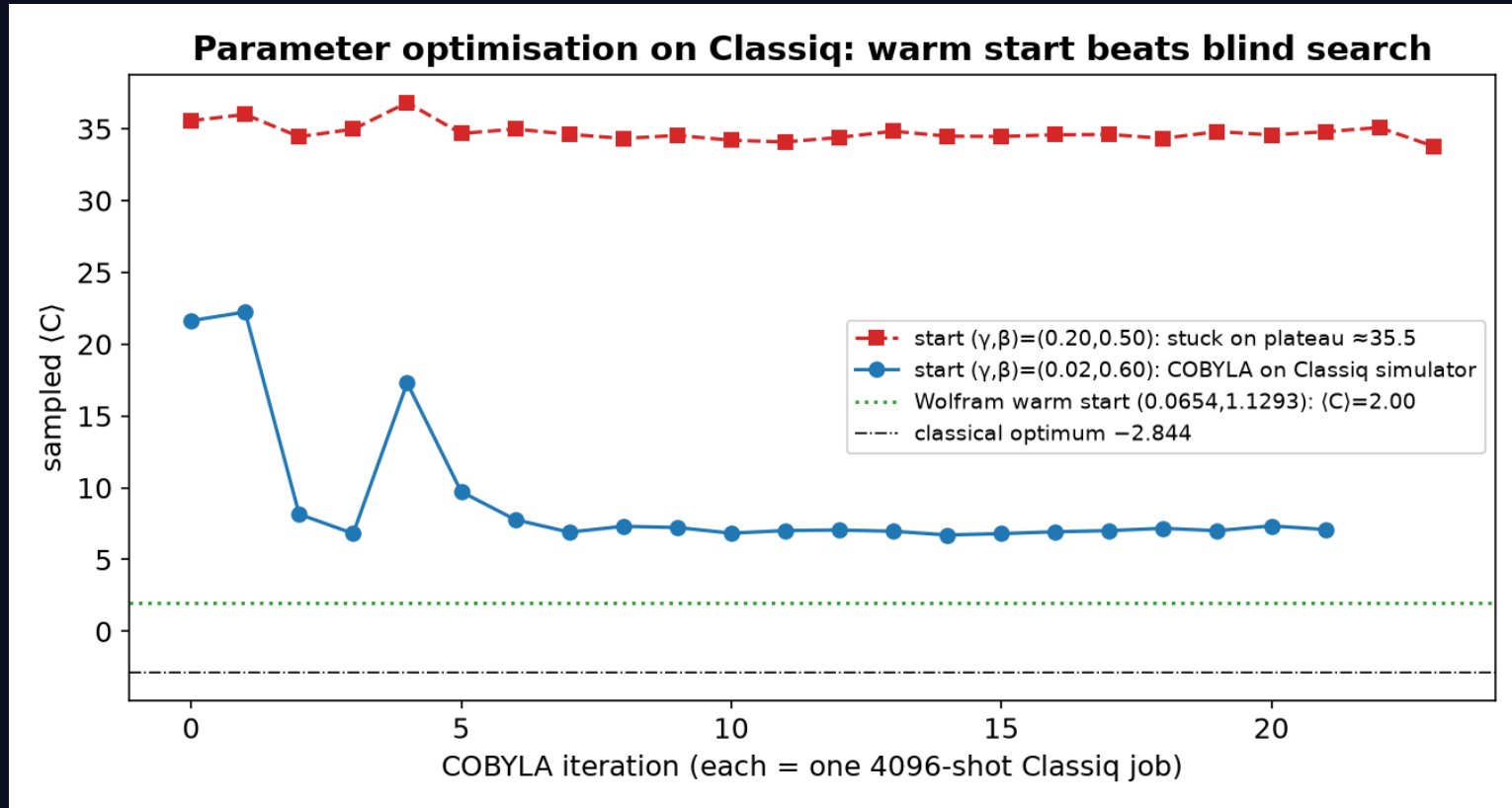
- ▶ Best sampled bitstring in every run = the classical optimum (0,3,0,0,0,0)
- ▶ p = 2: $\langle C \rangle = -0.466$ sampled vs -0.383 exact, $P(\text{cost} < 0) = 0.668$ vs 0.67
- ▶ Results, QASM and per-iteration traces saved in group/classiq/results/
- ▶ Demo video: group/video/qaoa_distribution.mp4 (distribution sharpening as angles ramp)

Demo: the QAOA distribution sharpening



- ▶ 120 frames, angles ramp from 0 to the $p = 2$ optimum
- ▶ Scatter: probability of each of the 4096 plans vs its cost
- ▶ Bars: $P(\text{cost} < 0)$ climbing to 0.67
- ▶ Black line: classical optimum -2.844
- ▶ Second clip:
group/video/classiq_convergence.mp4 —
COBYLA on Classiq, iteration by iteration

Optimising on Classiq: warm start beats blind search



- ▶ Start $(0.20, 0.50)$: COBYLA stuck at $\langle C \rangle \approx 35.5$ — the plateau Wolfram predicted
- ▶ Start $(0.02, 0.60)$: 20 Classiq jobs, $\langle C \rangle 21.6 \rightarrow 7.0$, still above warm start 2.0
- ▶ Shot noise (4096 shots) hides gradients; 15 parallel jobs let us sweep starts instead
- ▶ Takeaway: exact classical pre-simulation of small instances gives angles that transfer

Click to play: 20 COBYLA iterations on the Classiq simulator, one 4096-shot job each (11 s).

Lean 4 certificates: the classical facts are machine-checked

```

theorem decodeTilt_bijective : Function.Bijective (fun q : Fin 2 × Fin 2 => decodeTilt q.1 q.2) -- 2 qubits ↔ 4 tilts
theorem costS_optTilt       : costS optTilt = -28440 -- cost ×104 (decide)
theorem costS_lower_bound   : ∀ t : Fin 6 → Fin 4, -28440 ≤ costS t -- 4096 cases (native_decide)
theorem costS_unique        : ∀ t : Fin 6 → Fin 4, costS t = -28440 → t = optTilt -- optimum is unique
theorem cost_phase_preserves_probability : ∀ γ c a, ‖exp(γ c i) · a‖2 = ‖a‖2 -- phase layer keeps P
theorem two_smul_P0 / two_smul_P1 : 2•P0 = 1 + Z, 2•P1 = 1 - Z -- qubit → Pauli-Z

```

0 sorry

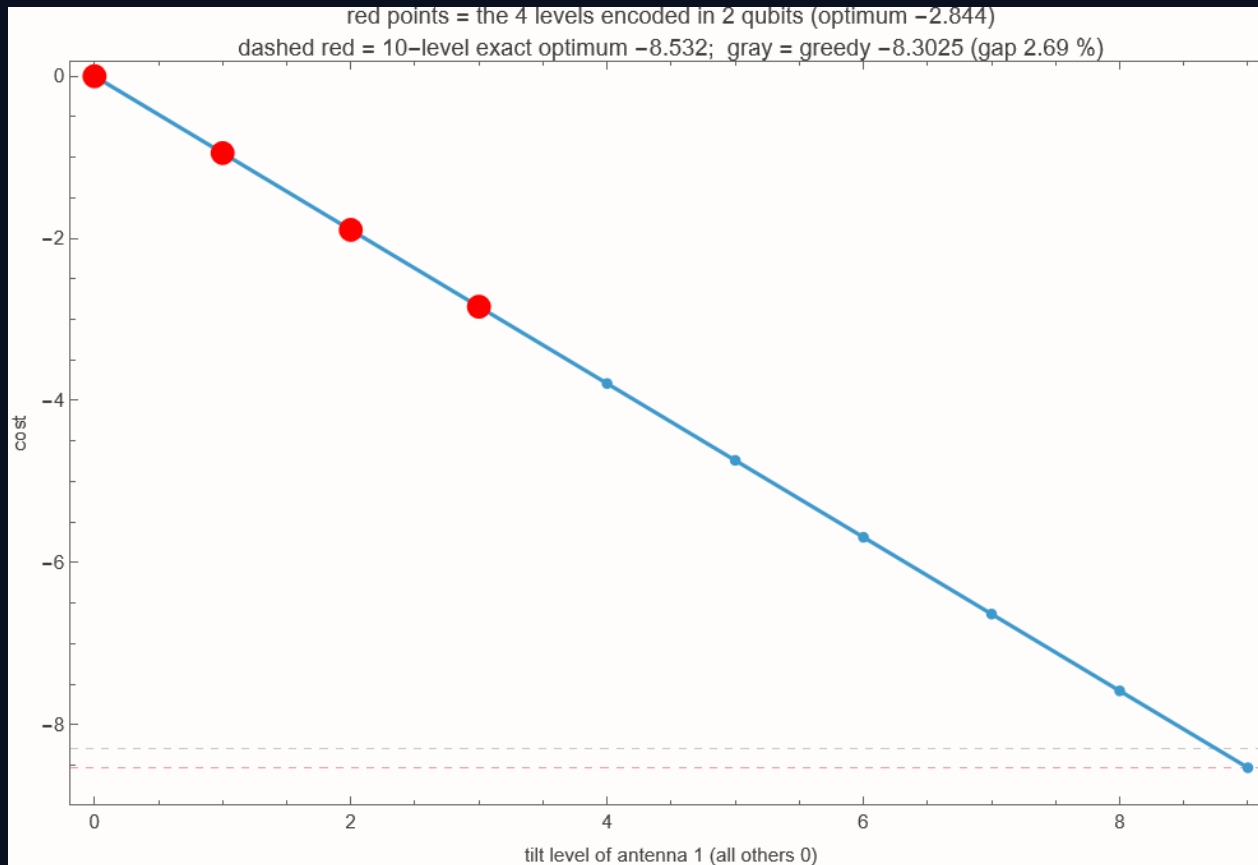
lake build succeeded

Mathlib

Lean 4.33.1

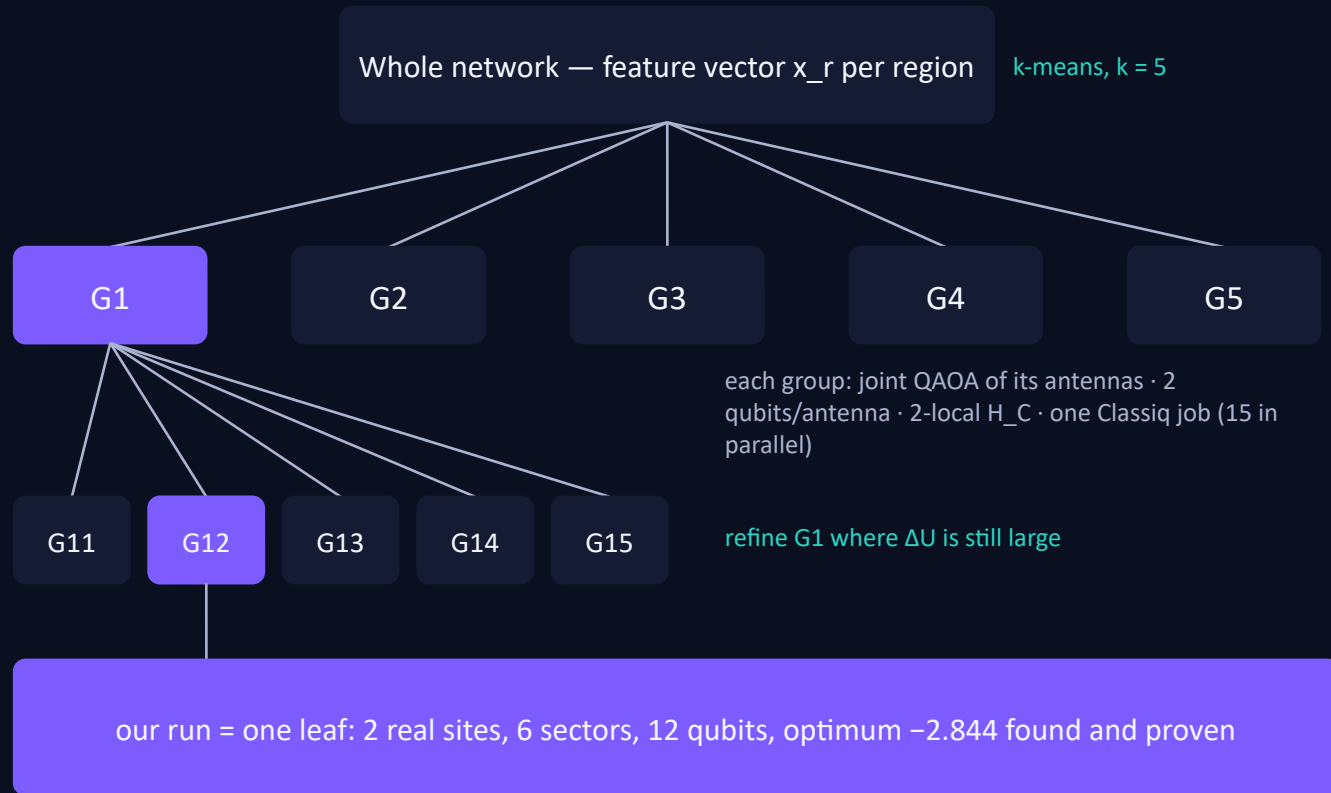
- ▶ group/lean/TiltQAOA.lean + build log
- ▶ Proves the optimum, its uniqueness and the encoding are exactly what the quantum runs report
- ▶ Quantum layer: unitarity of the phase operator, so QAOA never destroys probability mass

From 4 tilt levels to the operator's 10 — and beyond



- ▶ 10 levels $\rightarrow$ 4 qubits per antenna $\rightarrow$ 24 qubits for the same 6 antennas
- ▶ Hamiltonian stays 2-local: gate count grows quadratically, not exponentially
- ▶ 100 antennas $\times$ 10 levels $\approx$ 400 qubits — QAOA depth stays $p \cdot O(n^2)$
- ▶ Grover alternative: quadratic speed-up, needs a fault-tolerant machine and an oracle for “cost < threshold”; shown on a 16-state slice in the notebook
- ▶ Next: run $p = 1$ on IonQ simulator backend via Classiq, then hardware

Whole network: coarse-to-fine k-means, one QAOA job per group



Two stated assumptions

1. Regions in one group are similar enough to share the coarse tilt plan — refinement relaxes this level by level.
2. Interference between groups is ignored at each level — fixed by re-optimising boundary antennas with neighbours held fixed.

- ▶ Group = antennas optimised jointly (the coupled ZZ problem) — that is where QAOA does work; a single shared 27-way choice would be brute-forced classically
- ▶ Quantum budget per job stays what we ran today: 12 qubits = 6 antennas; 24 qubits = 6 antennas $\times$ 10 tilt levels
- ▶ Classical: k-means + utility aggregation; quantum: one warm-started QAOA per group

$5 \rightarrow 25 \rightarrow 125$ groups: distinct plans grow, jobs per level $\leq 25 \rightarrow$ two batches of 15

Everything is in the group folder — reproducible end to end

Artifact	Path (C:\Projects\Hackton\group\...)	Key number
Classiq QAOA program + results	classiq\qaoa_tilt_classiq.py, classiq\results*.json/*.qasm	$\langle C \rangle$ 2.020 (p1), -0.466 (p2)
Pauli decomposition + QASM layer	gates\pauli_terms.json, gates\qaoa_layer_schedule.qasm	73 terms, 2-local
Wolfram notebook, PDF, CDF, MD	wolfram\wolfram_realdataset_algorithms.*	$P(\text{cost} < 0)$ 0.024→0.53→0.67
Lean 4 proof + build log	lean\TiltQAOA.lean	optimum -2.844 unique, 0 sorry
Charts + demo videos	charts*.png, video*.mp4	120 + 22 frames
Hamiltonian write-up	qaoa_antenna_hamiltonian_explanation.md	—

Team Quantum Headache — thank you. Questions?